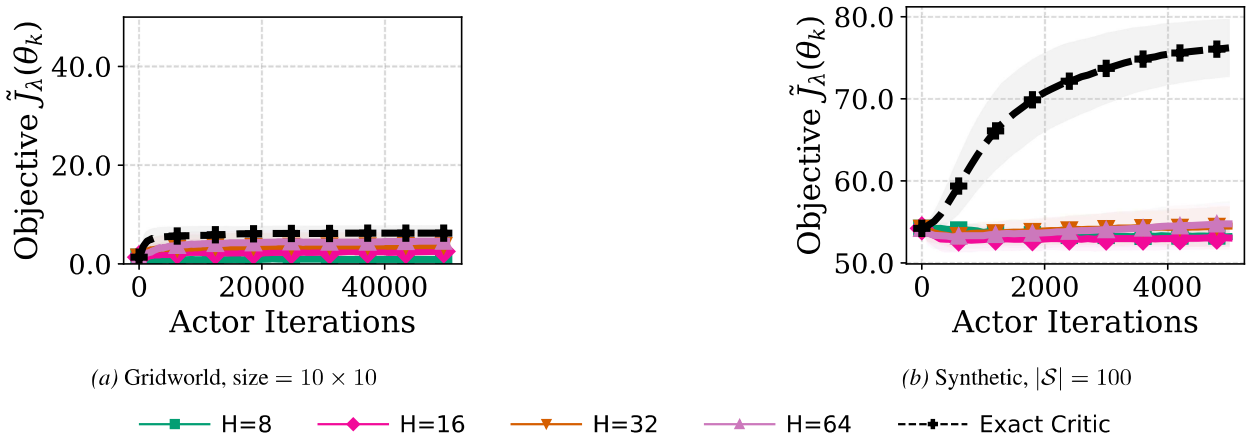

(a) Gridworld, size $= 10 \times 10$

(b) Synthetic, $|\mathcal{S}| = 100$

Figure 2. **Performance of `Ent-AC` across Gridworld and Synthetic environments.** We report the mean objective value $\tilde{J}_\lambda(\theta_k)$ as a function of actor iterations for varying critic update frequencies $H \in \{8, 16, 32, 64\}$. Panel (a) show results for tabular Gridworld of size 10×10, while panel (b) illustrate performance on synthetic MDPs with state space sizes $|\mathcal{S}| = 100$ and fixed action space of size $|\mathcal{A}| = 4$. The "Exact Critic" (dashed black line) represents the performance upper bound achieved by solving the critic to optimality. Shaded regions denote one standard deviation across 50 independent random seeds. Increasing H consistently reduces the approximation gap relative to the exact critic, which greatly enhances the performance of the algorithm.